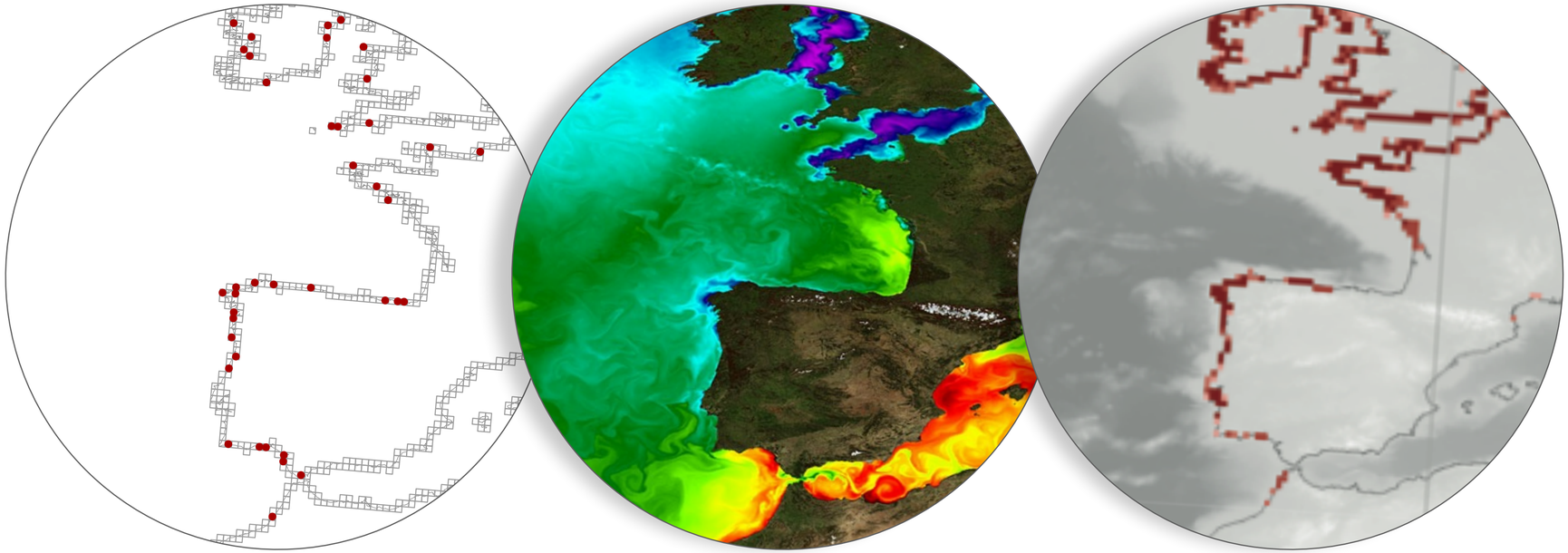
# Session 3: The Architecture of Species Distribution Models

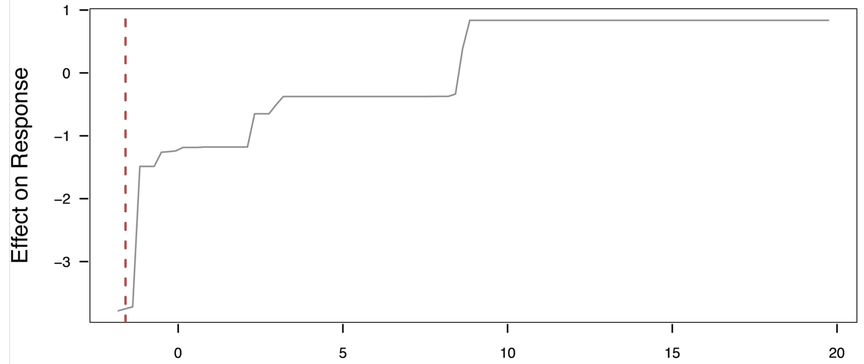
Modeling the distribution of biodiversity under global climate change

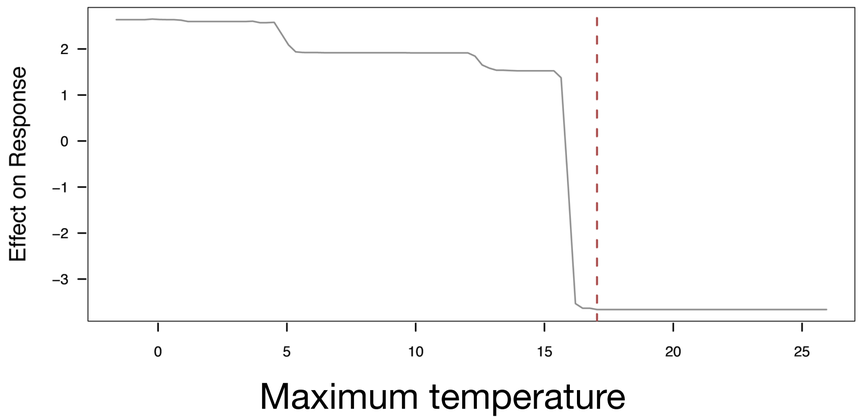
Jorge Assis, PhD // Nord University, Norway // biodiversityDS, Centre of Marine Sciences, Portugal

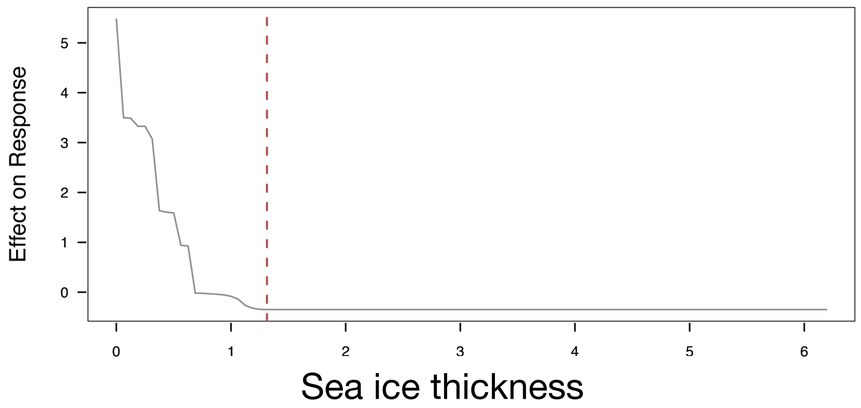
# Correlative SDMs

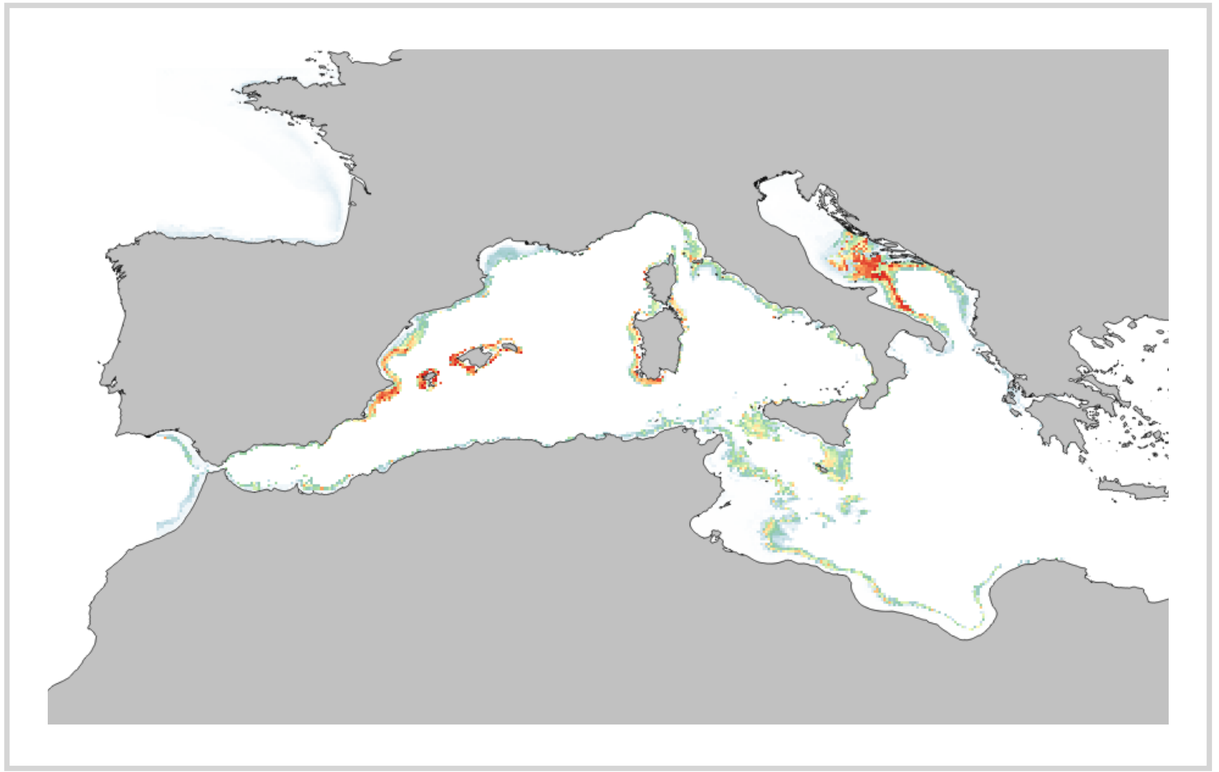


**Species distribution modeling** use **machine learning algorithms** linking **empirical occurrence records** with corresponding **environmental information** to explain and predict species habitat suitability.

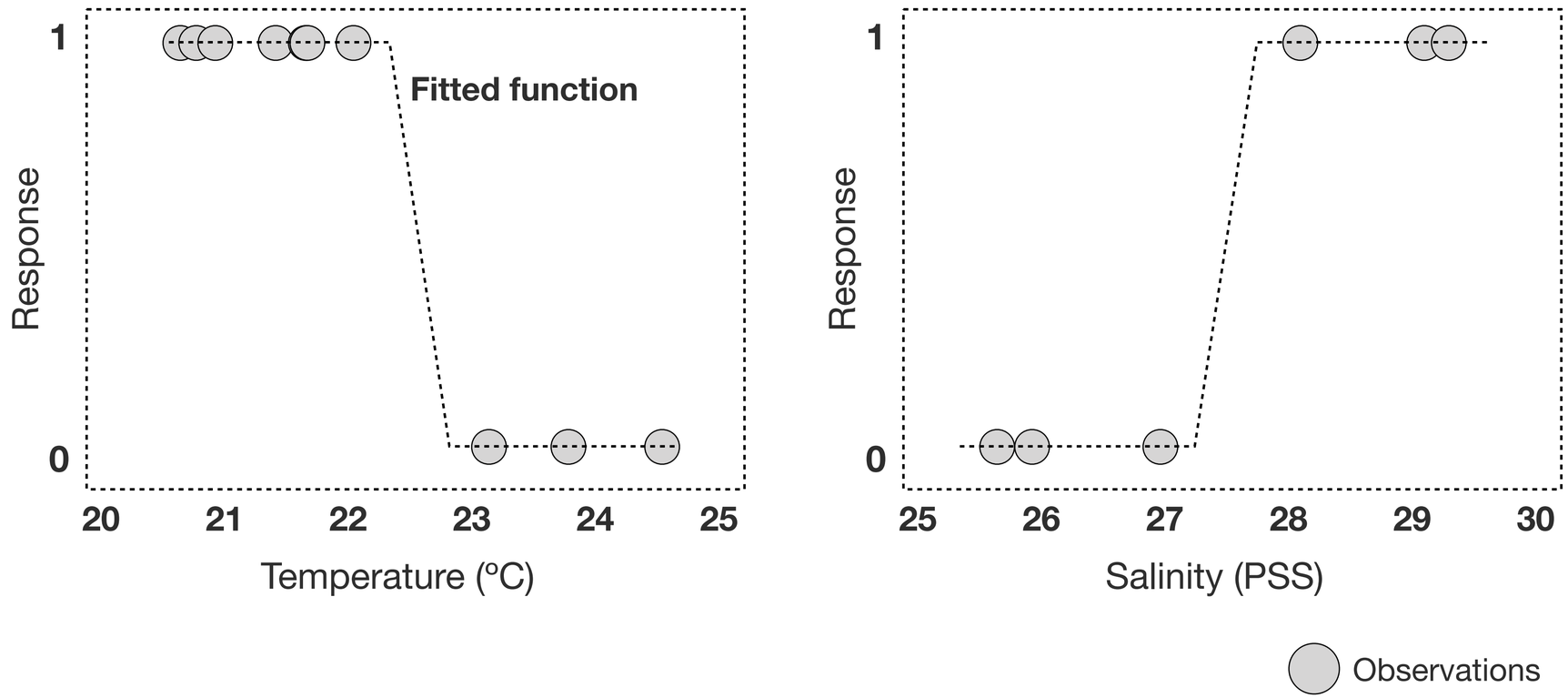


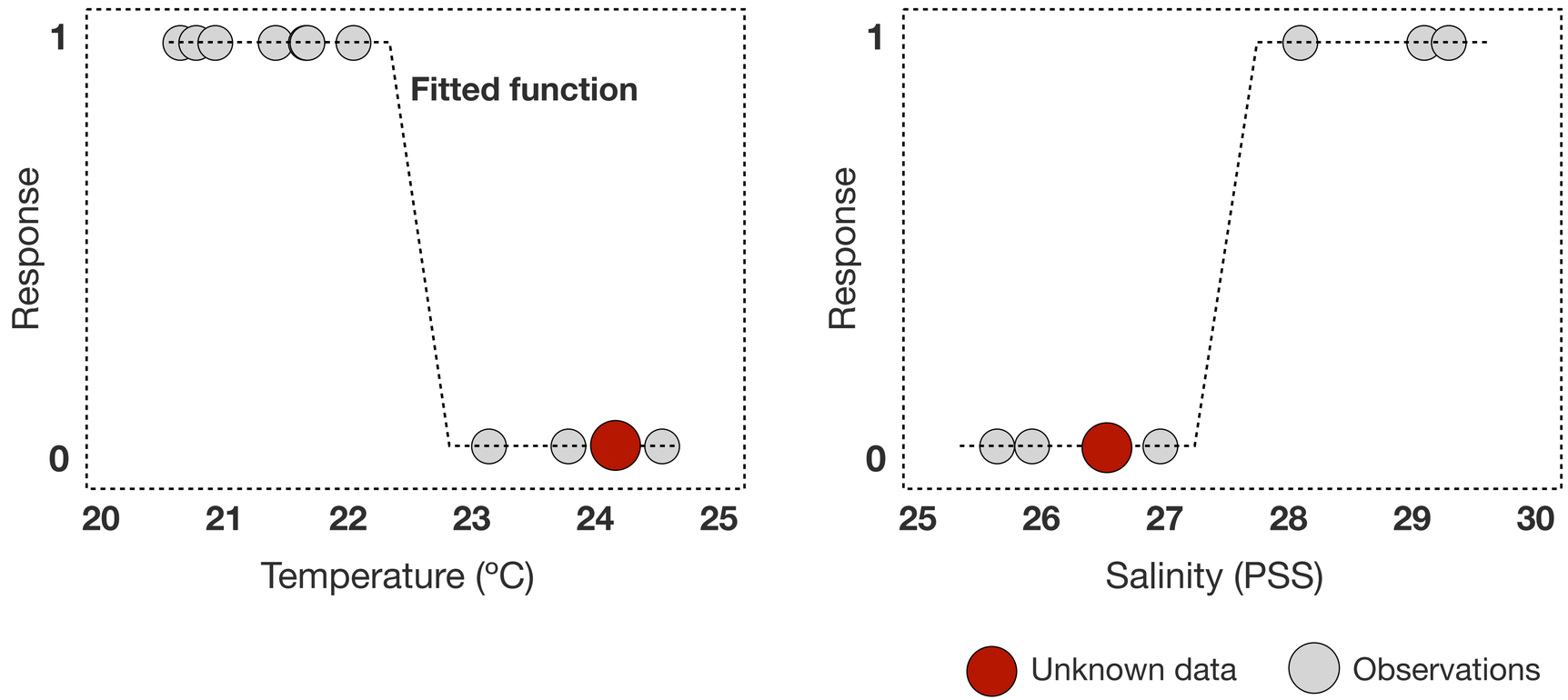



Machine learning (1) captures **the effect of multiple environmental variables** on species distributions; and (2) **predicts species occurrences** as a probability function across space and time (0-1).




# Model prediction

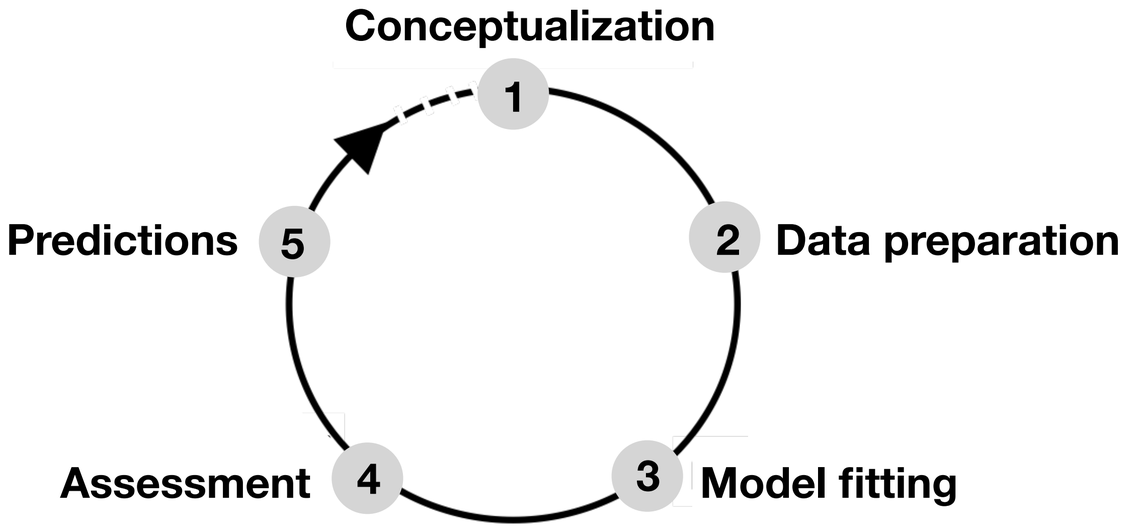
If a model can explain the relationship between distribution records and environmental variables, we can make predictions to unknown samples.




# Model prediction

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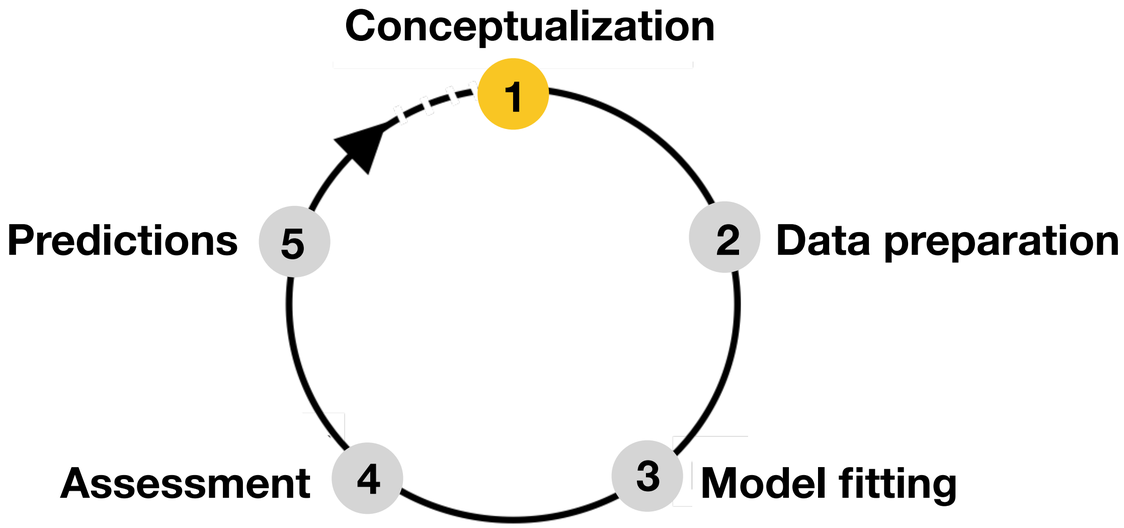
***e.g.,*** *Temp. = 24.5ºC & Sal. = 26.5 PSS, response is probability = 0 (i.e., absence).*




# Model building: an iterative loop

**Feedback loops:** building an SDM is not a linear process. It involves distinct steps, **with insights from later steps requiring revisiting earlier ones** *(e.g., improve biodiversity data collection or remove surplus environmental layers).*

The **goal is to build ecologically meaningful models for the purpose** *(e.g., conservation planning and climate impact assessment).*

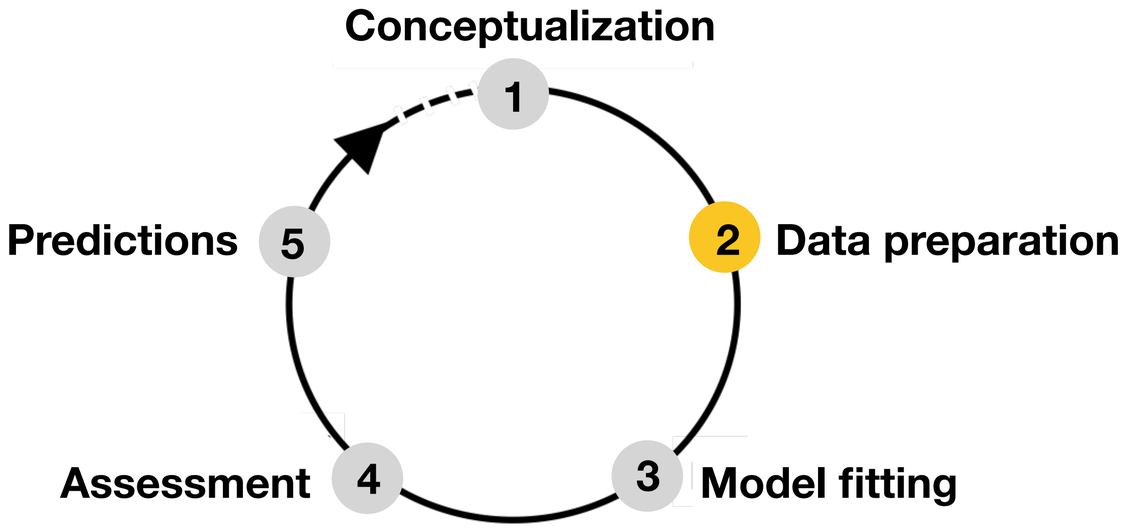



# Conceptualization

**Define the objective:** what question is the model trying to answer?
*(e.g., forecast species' distribution under climate change, identify invasion risk)*

**Get knowledge on the species:** about its distribution limits (literature).

**Hypothesize drivers:** What variables may influence its distribution? (literature).

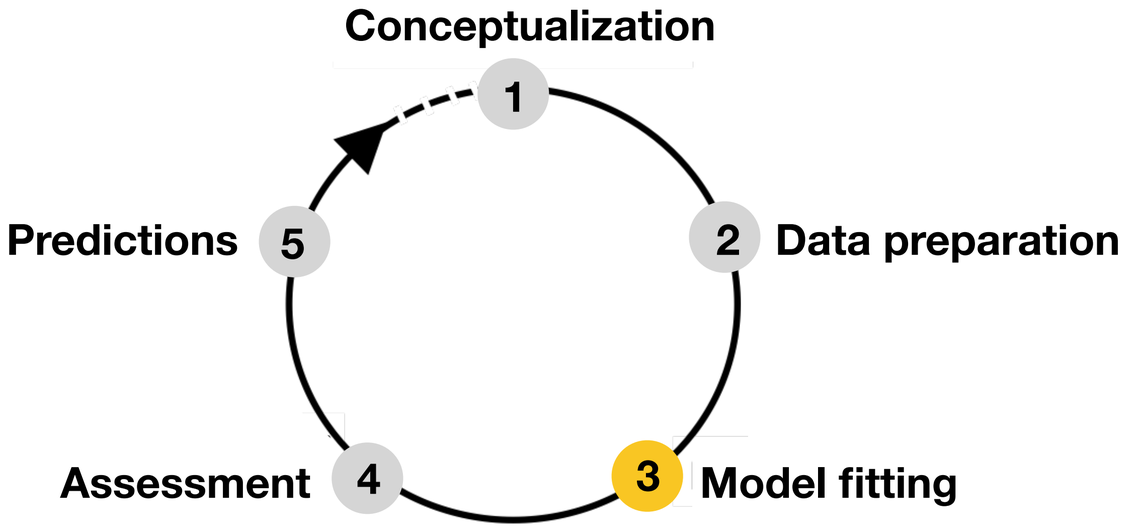



# Data preparation

**Gather and process species occurrence data** (e.g., check for taxonomic errors);

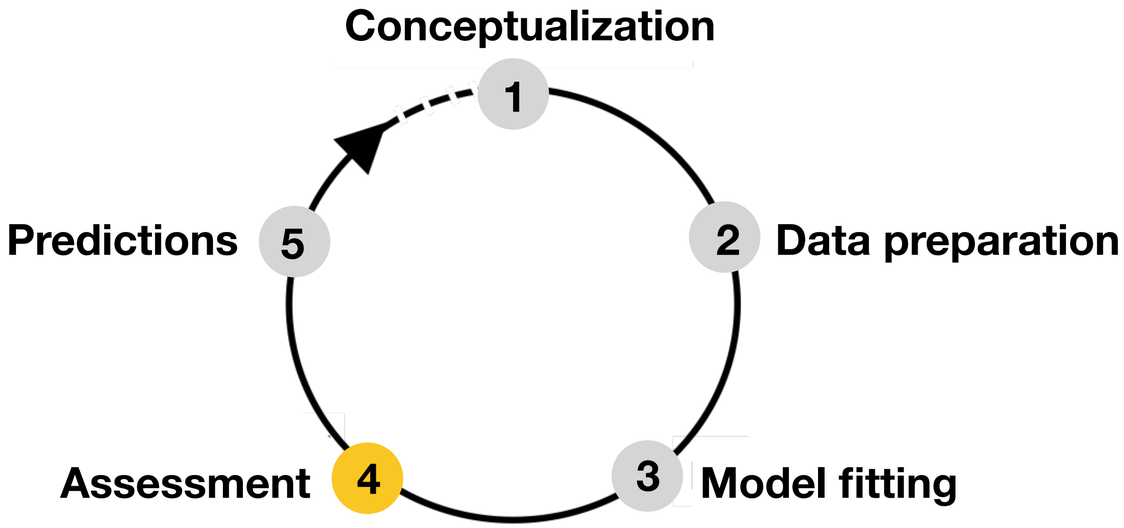
**Gather and process environmental variables relevant to the hypotheses**.

**Define the study area:** what geographic extent and spatial / temporal resolution are appropriate for the research question and data?




# Model fitting

**Algorithm selection:** choose a statistical method and the algorithm-specific parameters (e.g., regularization in MaxEnt, tree complexity in BRT).



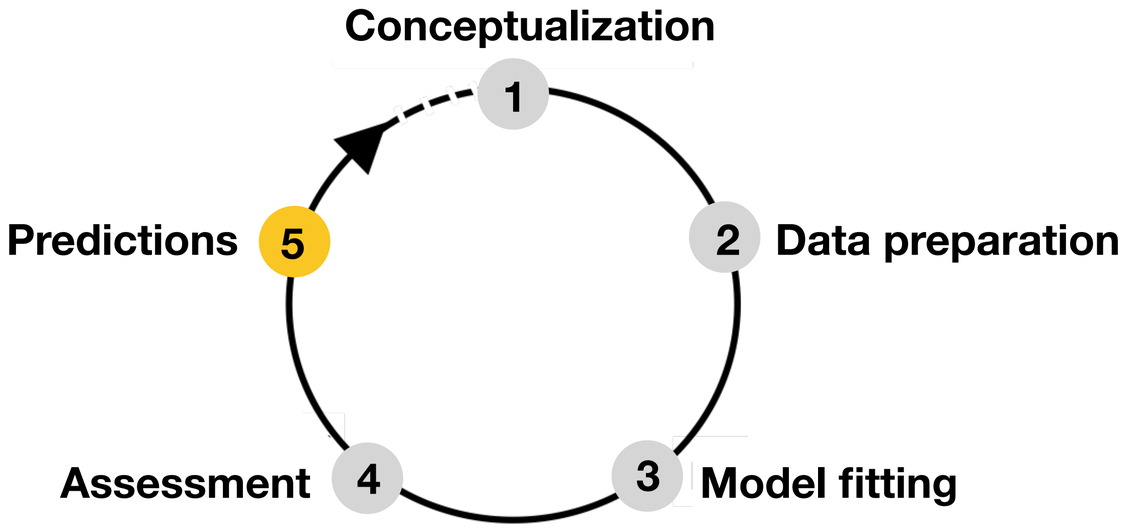

# Assessment of the model

## How good is the model?

Evaluating **performance**, as well as **ecological realism** of the model.

*Does the model accurately distinguish presence sites?*

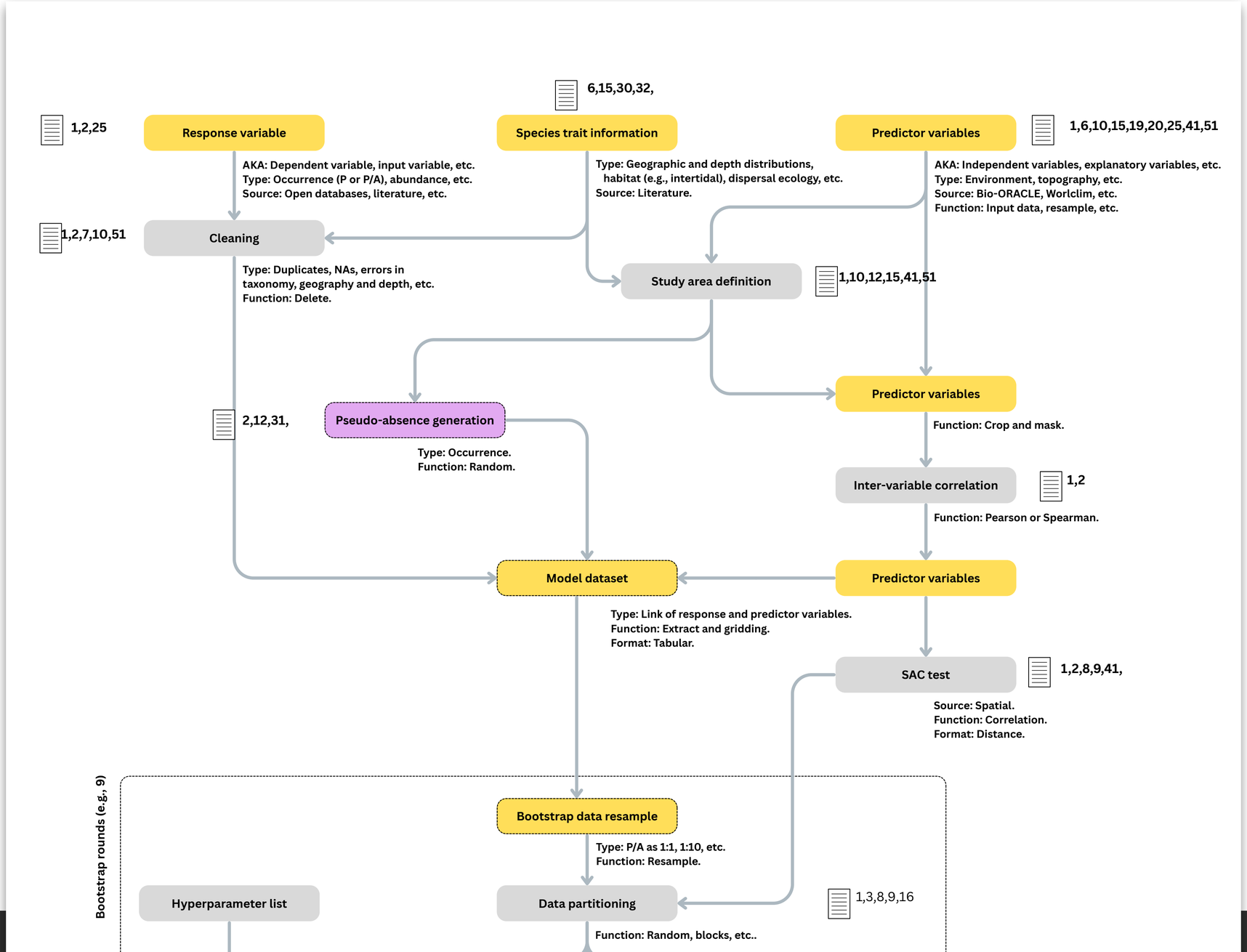
*Do the modeled relationships make biological sense? (ecological realism check)*

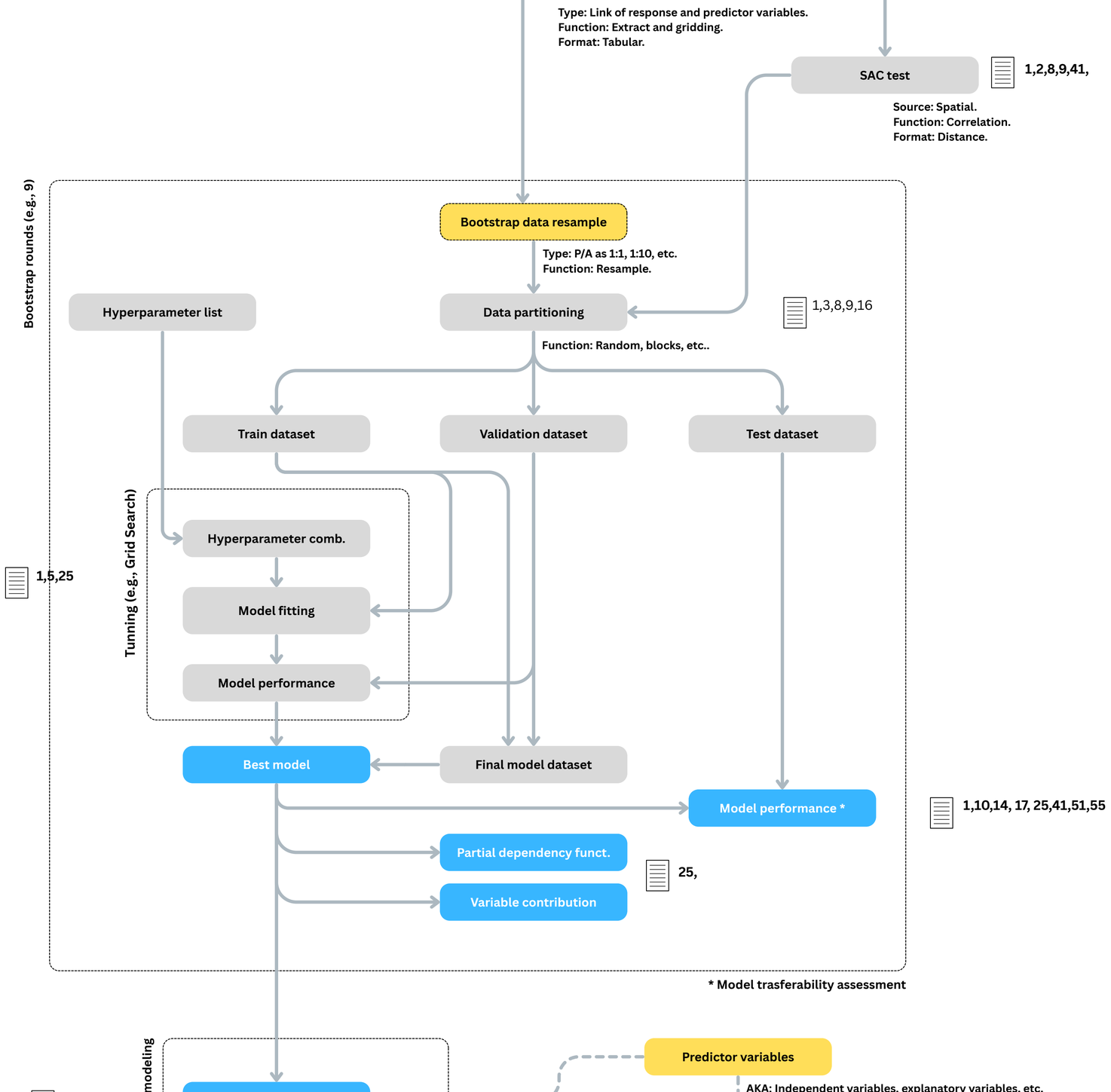


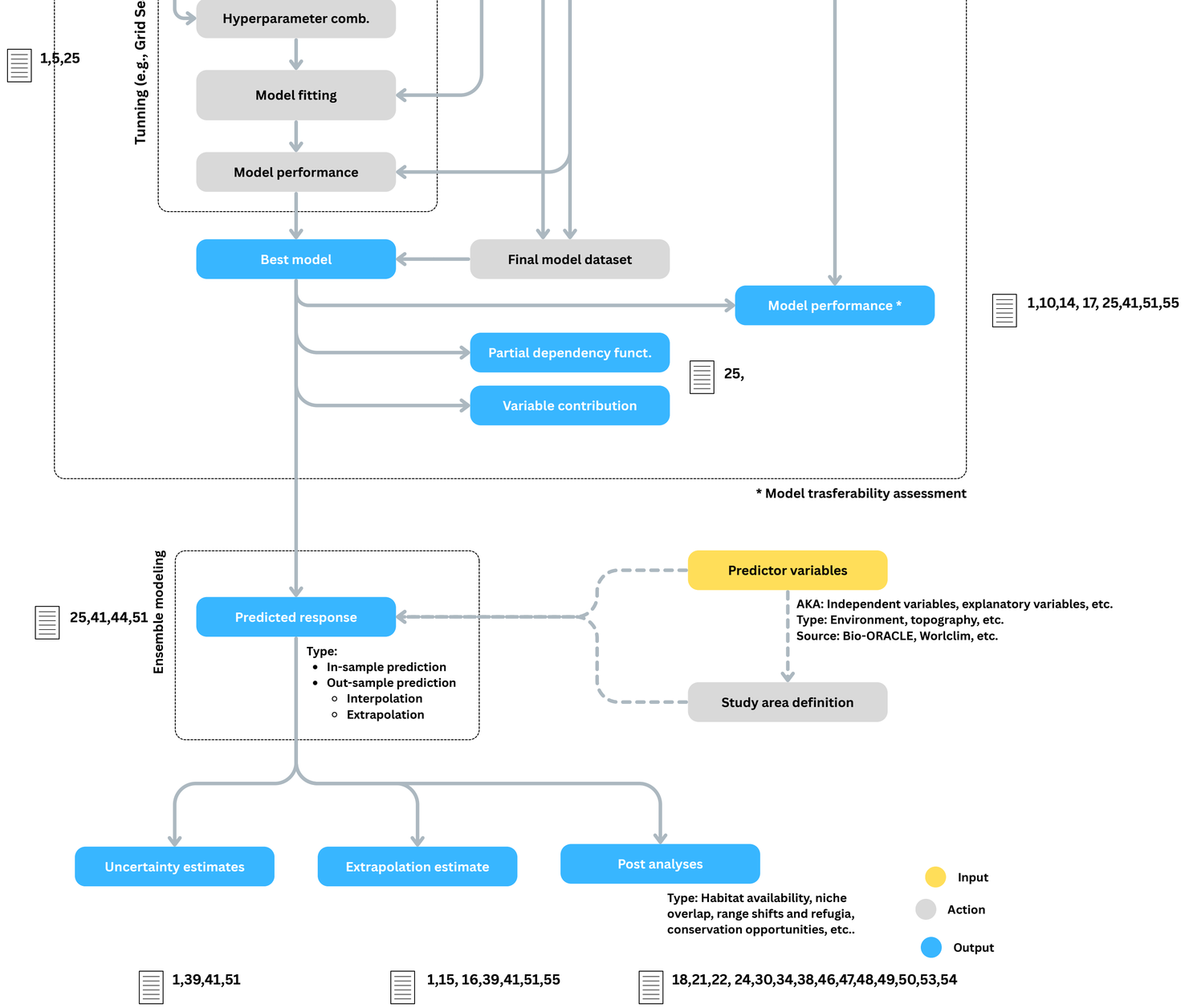

# Predictions in space and time

**Process:** a **model fits** observed species occurrence data against environmental variables for the corresponding period. The same model is then **applied directly to new climate data representing new environmental conditions**.

**Application:** predictions address the initial objective.
*(e.g., map future range shifts, assess invasion risk, identify conservation priorities).*


6,15,30,32,
1,2,25
Response variable
Species trait information
Predictor variables
1,6,10,15,19,20,25,41,51
AKA: Dependent variable, input variable, etc.
Type: Occurrence (P or P/A), abundance, etc.
Source: Open databases, literature, etc.
Type: Geographic and depth distributions, habitat (e.g., intertidal), dispersal ecology, etc.
Source: Literature.
AKA: Independent variables, explanatory variables, etc.
Type: Environment, topography, etc.
Source: Bio-ORACLE, Worlclim, etc.
Function: Input data, resample, etc.
1,2,7,10,51
Cleaning
Type: Duplicates, NAs, errors in taxonomy, geography and depth, etc.
Function: Delete.
Study area definition
1,10,12,15,41,51
Predictor variables
Function: Crop and mask.
2,12,31,
Pseudo-absence generation
Type: Occurrence.
Function: Random.
Inter-variable correlation
1,2
Function: Pearson or Spearman.
Model dataset
Predictor variables
Type: Link of response and predictor variables.
Function: Extract and gridding.
Format: Tabular.
SAC test
1,2,8,9,41,
Source: Spatial.
Function: Correlation.
Format: Distance.
Bootstrap rounds (e.g., 9)
Bootstrap data resample
Type: P/A as 1:1, 1:10, etc.
Function: Resample.
Hyperparameter list
Data partitioning
1,3,8,9,16
Function: Random, blocks, etc..


Type: Link of response and predictor variables.
Function: Extract and gridding.
Format: Tabular.
SAC test
1,2,8,9,41,
Source: Spatial.
Function: Correlation.
Format: Distance.
Bootstrap rounds (e.g., 9)
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Type: P/A as 1:1, 1:10, etc.
Function: Resample.
Hyperparameter list
Data partitioning
1,3,8,9,16
Function: Random, blocks, etc..
Train dataset
Validation dataset
Test dataset
Tunning (e.g., Grid Search)
Hyperparameter comb.
1,5,25
Model fitting
Model performance
Best model
Final model dataset
Model performance *
1,10,14, 17, 25,41,51,55
Partial dependency funct.
Variable contribution
25,
* Model trasferability assessment
modeling
Predictor variables
AKA: Independent variables, explanatory variables, etc.


Tunning (e.g., Grid Se
Hyperparameter comb.
1,5,25
Model fitting
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Model performance *
1,10,14, 17, 25,41,51,55
Partial dependency funct.
Variable contribution
25,
* Model trasferability assessment
Ensemble modeling
25,41,44,51
Predicted response
Type:
In-sample prediction
Out-sample prediction
Interpolation
Extrapolation
Predictor variables
AKA: Independent variables, explanatory variables, etc.
Type: Environment, topography, etc.
Source: Bio-ORACLE, Worlclim, etc.
Study area definition
Uncertainty estimates
Extrapolation estimate
Post analyses
Type: Habitat availability, niche overlap, range shifts and refugia, conservation opportunities, etc..
Input
Action
Output
1,39,41,51
1,15, 16,39,41,51,55
18,21,22, 24,30,34,38,46,47,48,49,50,53,54